

CHELFF, Algeria

WMO: 604250

Lat: 36.213N

Lon: 1.332E

Elev: 141

StdP: 99.64

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.8	(c) 4.9	(d) -0.9	(e) 3.6	(f) 15.3	(g) 0.6	(h) 4.0	(i) 13.0	(j) 14.1	(k) 14.1	(l) 12.5	(m) 13.5	(n) 1.5	(o) 60	(p) 0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.4	(c) 41.1	(d) 22.1	(e) 39.2	(f) 22.1	(g) 37.8	(h) 21.7	(i) 25.3	(j) 35.5	(k) 24.3	(l) 34.1	(m) 23.5	(n) 33.1	(o) 4.2	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.6	17.6	29.9	21.4	16.4	28.8	20.6	15.5	28.3	78.1	35.2	73.8	34.5	70.5	33.2	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.5	(b) 9.0	(c) 7.8	(e) 1.2	(f) 44.2	(g) 1.2	(h) 1.2	(i) 0.4	(j) 45.1	(k) -0.4	(l) 45.8	(m) -1.1	(n) 46.5	(o) -2.0	(p) 47.4		
(a) 10.5	(b) 9.0	(c) 7.8	(e) 0.8	(f) 26.4	(g) 1.0	(h) 2.1	(i) 0.1	(j) 27.9	(k) -0.5	(l) 29.2	(m) -1.1	(n) 30.3	(o) -1.8	(p) 31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	11.2	12.1	14.8	17.2	21.9	26.8	30.5	30.4	26.3	22.0	15.6	12.5
	DBStd	7.47	2.31	2.91	2.85	3.30	3.67	3.57	2.56	2.38	2.58	3.33	3.19	2.52
	HDD10.0	34	13	12	2	0	0	0	0	0	0	0	1	7
	HDD18.3	861	220	176	114	59	9	0	0	0	0	7	94	182
	CDD10.0	3747	52	70	151	217	368	505	637	633	489	373	168	84
	CDD18.3	1532	0	1	4	27	119	256	378	374	239	122	11	1
	CDH23.3	18362	0	3	52	276	1212	3106	5270	5075	2367	937	63	1
CDH26.7	10153	0	1	6	77	536	1744	3270	3052	1117	339	11	0	
Wind	WSAvg	2.9	2.8	3.1	3.0	3.1	3.1	3.1	3.0	2.8	2.7	2.5	2.7	2.5
Precipitation	PrecAvg	425	50	54	42	44	31	9	1	5	15	39	55	54
	PrecMax	641	178	141	126	138	98	50	14	41	74	132	172	129
	PrecMin	256	0	0	0	0	0	0	0	0	0	0	0	2
	PrecStd	113	34	41	29	36	28	11	3	9	17	35	39	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	23.2	27.2	31.9	38.1	41.7	44.1	43.0	38.8	35.0	28.1	22.2
		MCWB	14.1	14.3	15.7	19.6	21.6	21.2	22.4	23.0	22.5	19.7	18.1	15.0
	2%	DB	18.4	20.4	24.9	28.8	34.2	39.1	41.3	40.7	35.9	32.2	25.0	19.9
		MCWB	12.9	13.6	15.3	17.8	19.7	20.9	22.7	21.9	21.5	19.2	16.3	13.8
	5%	DB	16.9	18.9	22.6	26.5	31.9	37.1	39.6	39.0	34.2	30.1	22.8	18.2
		MCWB	12.3	13.1	14.5	16.9	18.9	21.0	22.3	22.4	21.2	18.7	15.7	13.1
	10%	DB	15.4	17.2	20.8	24.1	29.6	35.0	37.9	37.2	32.3	28.2	20.9	16.8
		MCWB	11.6	12.2	14.0	15.9	18.4	20.2	21.9	21.9	20.7	18.3	15.2	12.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	16.1	18.4	20.8	23.5	24.9	27.0	27.2	25.5	22.1	19.4	16.0
		MCDB	18.8	20.6	24.6	29.1	34.1	35.5	37.9	36.5	34.2	29.9	26.0	20.8
	2%	WB	13.6	14.7	16.6	18.9	21.3	23.2	24.9	25.7	23.7	20.8	17.7	14.7
		MCDB	17.2	18.9	22.0	26.6	30.9	34.6	36.3	35.6	31.2	28.5	22.4	18.7
	5%	WB	12.8	13.6	15.5	17.6	20.1	22.2	23.9	24.6	22.7	19.9	16.6	13.8
		MCDB	16.0	17.6	20.7	24.9	29.3	33.7	34.9	34.0	30.4	27.4	21.1	17.1
	10%	WB	11.9	12.7	14.5	16.5	19.1	21.2	23.2	23.7	22.0	19.2	15.6	13.0
		MCDB	15.0	16.4	19.5	22.9	27.8	31.7	33.9	32.7	29.3	26.2	19.9	15.9
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.5	10.7	11.7	12.6	13.8	14.4	13.7	12.1	11.2	8.7	8.4
		MCDBR	10.4	12.0	13.9	15.5	16.1	16.4	16.5	15.8	14.4	13.7	11.5	10.2
		MCWBR	6.2	6.5	7.0	7.2	6.4	5.5	5.8	5.4	5.1	5.6	5.7	
	5% WB	MCDBR	9.1	10.4	11.8	14.0	14.4	14.9	14.9	13.6	12.4	11.7	10.2	9.0
		MCWBR	5.8	6.2	6.8	7.3	6.8	5.9	6.1	5.6	5.2	4.8	5.6	5.6
Clear-Sky Solar Irradiance	taub	0.325	0.346	0.404	0.424	0.456	0.469	0.518	0.488	0.457	0.408	0.367	0.332	
	taud	2.489	2.424	2.237	2.203	2.136	2.114	1.971	2.062	2.163	2.296	2.391	2.484	
	Ebn at Noon	870	889	860	860	834	821	777	793	797	806	807	836	
	Edh at Noon	83	99	130	142	154	158	181	162	139	112	91	79	
All-Sky Solar Radiation	RadAvg	2.55	3.35	4.62	5.66	6.55	7.48	7.36	6.60	5.17	3.94	2.70	2.28	
	RadStd	0.26	0.37	0.35	0.44	0.50	0.28	0.16	0.27	0.25	0.28	0.23	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page